

PROJECT OVERVIEW

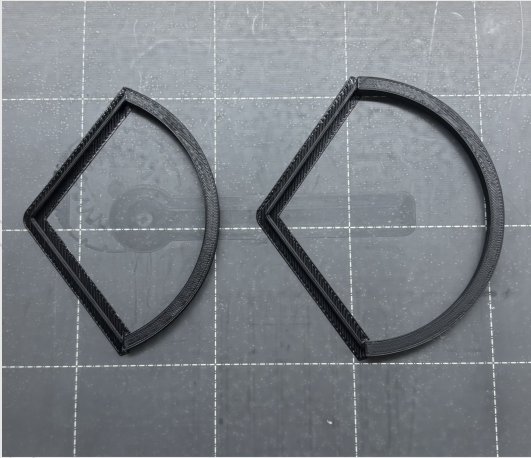
Question: Why are archery bows always tapered at the ends?

Goal: Create a mass-optimized bow with uniform stiffness across bow arms



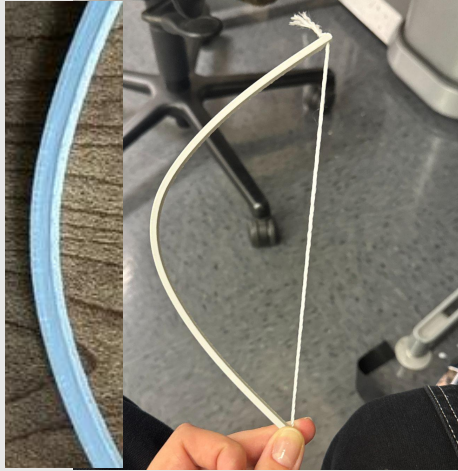
MANUFACTURING

Prototype 1



Initial project idea: analyzing various arm lengths to optimize draw force → too complex

Prototype 2



Changing project idea to focus on overall shape of bow in relation to the stress the force of the draw causes

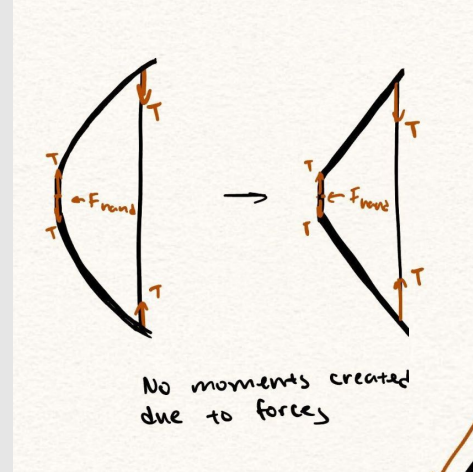
Final Model



Mass-optimized bow that distributes stress evenly across bow arms. Angled simplification is actually very accurate

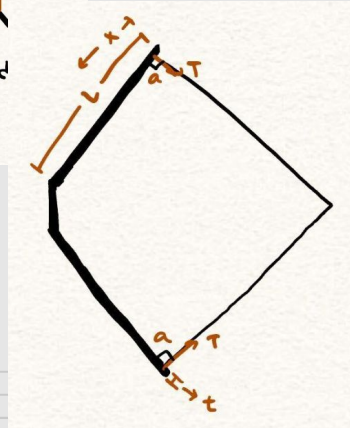
CALCULATIONS + SIMULATIONS

- Simplified bow arms into straight cantilevers
 - Accurate because of the way moments work
 - Analyzing cantilevers is more representative of what we've learned in this class
- Took cantilever beam equation and plugged in formulas concerning our scenario
- Decided to vary $b(x)$ as it makes more sense from a manufacturing standpoint and makes calculations easier.
- At $x=0$, we find the greatest width. As x approaches L , the width tapers linearly to 0
- Calculations uses a safety factor of 3 and scaled down 1:4 from a true adult-sized bow, according to our model



$$\sigma = \frac{M(x)y(x)}{I(x)}$$

$$b(x) = \frac{6T \sin(a)(L-x)}{\sigma t^2}$$





SIMULATIONS

- Kept simulations as close to reality as possible
 - Final model made with 100% infill PLA (yield stress 40 MPA), FEA simulations used acrylic, which has same yield stress
 - Split top and bottom arms in order to apply a constraint similar to a hand grip
- Expectations based off hand calcs:
 - F.S. of about 3
 - Stress of about 13.3 MPA
- Simulated several bow types for tapered and non-tapered bow arms:
 - Angled bow
 - Filleted bow
 - Straight bow

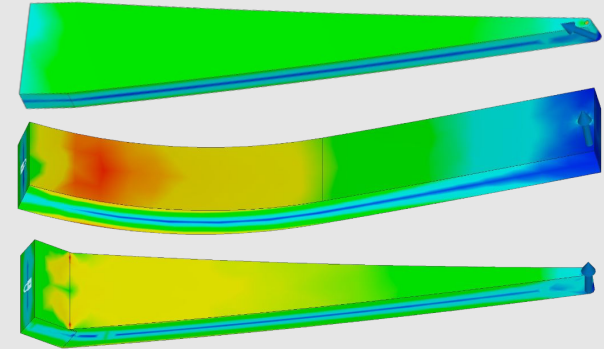


DISCUSSION + CONCLUSION

Hand calc results:

13.3 MPa of stress, 6 in bow arms, thickness of 1/8th in, solves to a width of 1.034 in, which tapers linearly to a width of 0 in over the 6 in bow arm

	Max. Von Mises (MPa)	Min. Factor of Safety
Taper Angled	16.315	2.452
Taper Straight	19.273	2.07
Taper Fillet	10.19	3.953
Rectangular Angled	12.437	3.216
Rectangular Straight	11.161	3.854
Rectangular Fillet	9.329	4.288



Future directions:

- Further analyze filleted bow—more mass can be reduced
- Look into different material properties to further reduce mass
- Look into bow handles/grips